

**AN INTELLIGENT YOUTUBE CONTENT FILTERING AND PERSONALISED
RECOMMENDATION SYSTEM FOR CHILDREN USING AGE-AWARE VIDEO
ANALYSIS**

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BONAFIDE CERTIFICATE

Certified that this project report titled **An Intelligent Youtube Content Filtering And Personalised Recommendation System For Children Using Age-aware Video Analysis** is the bonafide work of **SOUNDARYA K (2024178044)** who carried out the project work under my supervision. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents an intelligent video recommendation system designed specifically for children, focusing on safety, age-appropriateness, and content relevance. With the rapid increase in digital media consumption among young users, ensuring that children are exposed only to suitable and educational content has become a critical challenge. This system aims to address that issue using machine learning and content analysis techniques.

The proposed system integrates multiple modules including video ingestion, feature extraction, content classification, and recommendation generation. Videos are analyzed based on key attributes such as age suitability, topic category, difficulty level, and safety score. A supervised machine learning model is employed to learn from labeled data and predict these attributes for new, unseen videos. Additionally, speech-to-text techniques are used to extract and analyze video transcripts, enabling deeper semantic understanding of the content.

To generate personalized recommendations, the system uses vectorization and similarity-based approaches such as cosine similarity, ensuring that recommended videos align with the child's profile and previously consumed content. The architecture also incorporates a pre-analysis and caching mechanism to optimize performance and deliver real-time recommendations efficiently.

By combining machine learning, natural language processing, and intelligent ranking strategies, this project provides a scalable and effective solution for creating a safe digital environment for children. The system not only improves content filtering but also enhances engagement by recommending relevant and age-appropriate videos.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid growth of digital media platforms has significantly transformed the way children consume content. Among these platforms, YouTube has emerged as one of the most widely accessed sources of entertainment and education for children across different age groups. While YouTube offers a vast collection of educational videos, nursery rhymes, cartoons, and learning resources, it also hosts a large volume of unregulated and potentially inappropriate content. This creates a critical challenge in ensuring that children are exposed only to age-appropriate, safe, and meaningful videos.

Children between the ages of 1 and 12 represent a particularly vulnerable user group, as they lack the cognitive maturity to distinguish between suitable and harmful content. Although platforms such as YouTube Kids attempt to address this issue, several studies and real-world incidents have revealed limitations in automated moderation systems, including exposure to misleading, violent, or psychologically inappropriate material. These shortcomings highlight the need for a more intelligent, fine-grained, and explainable content filtering system that goes beyond simple keyword blocking or category-based restrictions.

A significant motivation for this research arises from the phenomenon commonly referred to as “Elsagate.” Elsagate describes a wave of videos that appeared on YouTube and similar platforms, where popular children’s characters such as those from animated films and nursery rhymes were used to depict disturbing, violent, sexualized, or psychologically inappropriate scenarios. These videos were often disguised using child-friendly thumbnails, titles, and tags, enabling them to bypass basic content moderation systems and appear in recommendations for young children. The incident exposed critical weaknesses in automated filtering mechanisms that relied heavily on metadata and user reports rather than contextual understanding of content. Despite subsequent platform-level interventions, studies and reports indicate that similar deceptive patterns continue to emerge, particularly through algorithmic exploitation and content farm strategies. Elsagate highlights the urgent need for age-aware, multi-layered content analysis systems that can evaluate semantic intent, repetition patterns, and developmental suitability rather than relying solely on surface-level indicators. Addressing such vulnerabilities is essential for ensuring psychological safety and trust in digital platforms designed for children.

Traditional content filtering mechanisms primarily rely on metadata such as video titles, user-reported tags, and channel-level categorization. However, such approaches are insufficient

due to noisy metadata, inconsistent labeling, and the dynamic nature of online content creation. Moreover, most existing systems fail to account for age-specific cognitive and emotional development, treating all children as a single homogeneous group. This lack of age awareness can result in either over-restriction—limiting educational exposure—or under-restriction, exposing children to unsuitable material.

Recent advances in machine learning, natural language processing, and representation learning provide an opportunity to address these challenges more effectively. By analyzing video transcripts, descriptions, tags, and contextual embeddings, intelligent systems can infer the semantic intent, educational value, and potential risk of a video. When combined with supervised learning and age-aware labeling strategies, such systems can dynamically classify content and personalize recommendations according to a child’s developmental stage.

This project proposes an Intelligent YouTube Content Filtering and Personalized Recommendation System for Children using Age-Aware Video Analysis. The system is designed to fetch videos selectively from trusted channels, analyze their textual content, assess appropriateness, classify topics, and assign age suitability labels. By integrating rule-based filtering with supervised machine learning models, the proposed framework ensures both precision and transparency in decision-making.

The system architecture emphasizes safety-first ingestion, where videos are fetched only from a curated set of approved channels, followed by multi-stage filtering that includes duplicate elimination, inappropriate content detection, topic classification, and age-bucket assignment. The processed videos are stored in a structured database and served through a controlled recommendation pipeline, ensuring that only validated content appears in the child-facing user interface.

By combining database-driven ingestion, machine learning-based classification, and age-aware recommendation logic, this project aims to provide a scalable, explainable, and reliable solution for children’s video consumption. The proposed system not only enhances content safety but also supports cognitive development by delivering relevant educational material tailored to specific age groups.

1.2 CHILDREN AND DIGITAL CONTENT CONSUMPTION

Digital platforms have become an integral part of early childhood learning and entertainment. Educational videos, animated rhymes, storytelling, and interactive content play a significant role in language development, behavioral learning, and cognitive growth. However, uncontrolled exposure to digital media can negatively impact attention span, emotional regulation, and behavioral patterns in children.

Children’s comprehension abilities vary significantly across age groups. For instance, toddlers (1–3 years) respond primarily to visual and auditory stimuli, while preschool and early school-age children (4–12 years) begin to engage with narrative structure, problem-solving concepts, and abstract ideas. A content recommendation system that does not differentiate between these developmental stages risks mismatching content complexity and intent.

Ensuring age-appropriate content delivery is therefore essential for promoting healthy digital engagement and minimizing potential psychological harm.

1.3 PROBLEM STATEMENT

Despite the availability of parental controls and platform-level moderation, children continue to be exposed to inappropriate, misleading, or developmentally unsuitable content on video-sharing platforms. Existing content filtering systems suffer from several limitations:

1. Over-reliance on unreliable metadata such as titles and tags.
2. Inadequate detection of implicit inappropriate themes.
3. Lack of fine-grained age classification.
4. Absence of explainable filtering decisions.
5. Inability to personalize recommendations based on developmental stages.

Furthermore, unrestricted video ingestion from open platforms leads to duplication, inconsistent labeling, and unsafe content propagation within recommendation pipelines. These challenges necessitate an intelligent, age-aware system capable of filtering, classifying, and recommending children’s videos with high accuracy and reliability.

1.4 PROPOSED SOLUTION

The proposed system introduces a structured, multi-layered pipeline for intelligent video filtering and recommendation. Videos are fetched exclusively from a predefined list of trusted YouTube channels to ensure source reliability.

A rule-assisted supervised learning approach is employed to label videos for appropriateness using keyword-based signals and category constraints. Videos deemed safe are further processed through a topic classification model that assigns semantic categories such as language learning, science, arts, or behavioral education.

Age-aware labeling is achieved by mapping content complexity, vocabulary usage, and semantic patterns to predefined age buckets, including nursery-level content. Only videos that

successfully pass all validation stages are stored with explicit age and topic labels and made available for recommendation.

This architecture ensures that unsafe or irrelevant videos are blocked early, while suitable content is accurately classified and delivered to the appropriate age group.

1.5 OBJECTIVES OF THE STUDY

The primary objectives of this project are:

1. To design a safe and controlled video ingestion pipeline using trusted channels.
2. To detect and block inappropriate content using rule-based and machine learning techniques.
3. To classify videos into meaningful educational topics.
4. To implement age-aware content labeling for children across different developmental stages.
5. To store validated videos in a structured database for reliable retrieval.
6. To support personalized and safe content delivery in child-facing applications.

1.6 SCOPE OF THE PROJECT

The scope of this project encompasses the design and implementation of an intelligent, multi-layered video content filtering and personalized recommendation system for children, with a strong emphasis on safety, age-appropriateness, and parental control. The system operates across multiple stages of content ingestion, analysis, classification, and presentation, ensuring that children are exposed only to developmentally suitable digital media.

The project includes an age-aware video analysis module that classifies content into distinct developmental groups, such as nursery (ages 1–3), early school (ages 4–8), and middle school (ages 9–12), using natural language processing techniques on video transcripts, titles, descriptions, and metadata. In addition, the system supports camera-based age estimation, enabling real-time age verification to prevent unauthorized access to content intended for older children.

The project further incorporates object and scene analysis within videos using computer vision techniques, enabling detection of visual elements that may be unsuitable for children, such as weapons, frightening imagery, or medical instruments. This visual analysis complements textual and metadata-based filtering, providing a more robust and comprehensive safety evaluation.

To ensure transparency and control, the system features a child-facing user interface designed for simplicity and accessibility, as well as a dedicated parental control interface. The parental interface allows guardians to monitor viewing history, manage age settings, review content recommendations, and impose additional restrictions based on individual preferences.

1.7 TOOLS AND TECHNOLOGIES

- Python for data processing and model development
- Supabase (PostgreSQL) for structured video storage
- YouTube Data API for controlled video ingestion
- Scikit-learn for supervised learning models
- Sentence Transformers for semantic embedding generation
- TF-IDF Vectorization for textual feature extraction
- YOLOV8 for Object Analysis in the videos.

1.8 ORGANISATION OF THE REPORT

This report is structured to present the proposed intelligent age-aware and safety-oriented video filtering and recommendation system for children in a clear and logical manner.

Chapter 2 provides a comprehensive literature survey of existing research related to children's content filtering, age classification, safety detection, and recommendation systems. It reviews prior studies, datasets, regulatory frameworks, and machine learning techniques that inspired and influenced the design of the proposed system, while identifying research gaps addressed by this project.

Chapter 3 describes the overall system architecture and design methodology. This chapter explains the functional components of the system, including video ingestion, age classification, safety and appropriateness analysis, topic classification, recommendation logic, and user interface design. The interaction between system modules and the data flow across different stages are also detailed.

Chapter 4 presents the implementation details of the proposed system along with experimental results. It discusses the tools, technologies, datasets, and machine learning models used, and explains how each component was implemented. Experimental outcomes obtained during age classification, safety filtering, and topic classification are also reported and analyzed.

Chapter 5 focuses on system evaluation and performance analysis. This chapter assesses the effectiveness of the proposed system using appropriate metrics such as accuracy, precision, recall, and confidence thresholds. It also analyzes system behavior in real-world scenarios, including false positives, blocked content cases, and recommendation reliability.

Chapter 6 concludes the report by summarizing the overall contributions and outcomes of the project. It includes the limitations of the current system and outlines possible future enhancements, including multimodal analysis, improved personalization, and advanced parental control mechanisms.

In summary, this report systematically documents the design, implementation, and evaluation of an intelligent age-aware and safety-oriented video filtering and recommendation system for children. The report progresses through architectural design, implementation, and performance evaluation, presenting a solution to the challenges of child safety and age-appropriate content in online video platforms. The structured organization ensures clarity, traceability, and ease of understanding for readers, researchers, and evaluators

CHAPTER 2

LITERATURE SURVEY

2.1 ONLINE VIDEO PLATFORMS AND CHILD SAFETY CONCERNS

The rapid adoption of video-sharing platforms among children has raised significant concerns regarding age-inappropriate and unsafe content exposure. Livingstone et al. [1] studied children’s interaction with digital media and concluded that algorithm-driven recommendation systems often prioritize engagement over safety, thereby increasing the likelihood of children encountering unsuitable content. This finding motivated the need for an external, child-centric filtering layer, independent of platform-level moderation.

Papadamou et al. [2] conducted one of the earliest systematic investigations into the *Elsagate* phenomenon, where videos appearing child-friendly contained disturbing, violent, or psychologically harmful narratives. Their study demonstrated that children could be led from benign videos to disturbing content within a few recommendation hops. This work strongly influenced the proposed system’s focus on content-level analysis rather than reliance on video titles, thumbnails, or channel reputation alone.

2.2 ANALYSIS OF DISTURBING AND HARMFUL CONTENT

Further expanding on Elsagate-related risks, Papadamou et al. [3] analyzed YouTube’s recommendation graph and showed that harmful children-targeted videos are densely interconnected, enabling rapid exposure escalation. These findings highlighted the need for proactive safety classification before recommendation, rather than post-consumption moderation.

Pavlopoulos et al. [4] proposed structured frameworks for identifying abusive and harmful online content, emphasizing that harm can be implicit, contextual, and narrative-driven. Similarly, Meta AI introduced the MetaHarm dataset [5], which categorizes harmful content into psychological, emotional, and implicit harm types. Although MetaHarm is not child-specific, its taxonomy informs the appropriateness and safety module of this project, particularly for detecting subtle disturbing patterns that evade keyword-based filters.

2.3 CHILD LANGUAGE RESOURCES AND AGE-SPECIFIC LINGUISTIC PATTERNS

MacWhinney et al. introduced the CHILDES (Child Language Data Exchange System) as part of the TalkBank project [6]. CHILDES provides transcribed child-directed and child-produced

speech annotated with age information. This dataset has been widely used to study linguistic development and age-dependent language structures.

The broader TalkBank spoken language corpora [7] extend this work by providing multilingual conversational datasets, enabling analysis of vocabulary growth, repetition, and sentence simplicity across age groups. These datasets inspired the project’s transcript-based age classification approach, particularly for distinguishing nursery, early-school, and middle-school language patterns.

2.4 TEXT-BASED AGE ESTIMATION AND THE TEXTAGE DATASET

Xia et al. proposed TextAge, a benchmark dataset and framework for predicting age groups from textual content [8]. Their study demonstrated that features such as lexical diversity, repetition frequency, and semantic concreteness vary significantly across age groups. This research validated the feasibility of text-only age prediction, directly motivating the use of video transcripts, titles, and descriptions in the proposed system.

Subsequent studies extended text-based age estimation to subtitles and speech transcripts [9], reinforcing the decision to prioritize language-based age analysis over visual-only approaches, especially for educational and nursery videos.

2.5 YOUTUBE METADATA AND TOPIC CLASSIFICATION

Large-scale YouTube analysis has been facilitated by datasets such as YouTube-8M, introduced by Abu-El-Haija et al. [10], which demonstrated the effectiveness of metadata and textual features for content categorization. Region-specific datasets such as the IN YouTube Trending Dataset [11] further revealed category distributions and topic patterns within real-world video data.

2.6 SPEECH-TO-TEXT FOR VIDEO UNDERSTANDING

Radford et al. introduced Whisper, a robust automatic speech recognition model capable of handling multilingual audio and varied accents [12]. Studies evaluating Whisper showed high transcription accuracy even for children’s speech, making it suitable for generating transcripts used in age, topic, and safety analysis within the proposed system.

2.7 REGULATORY FRAMEWORKS AND ETHICAL CONSIDERATIONS

The Children’s Online Privacy Protection Act (COPPA), enforced by the U.S. Federal Trade Commission [13], establishes legal requirements for handling children’s data online. Studies on COPPA compliance emphasize minimizing data collection, enabling parental oversight, and

ensuring transparency in recommendation systems [14]. These principles directly influenced the inclusion of a parental user interface, controlled recommendation logic, and cautious behavioral data usage in this project.

2.8 MACHINE LEARNING TECHNIQUES IN PRIOR WORK

Existing research on child-oriented content classification has employed both traditional and deep learning approaches. Classical machine learning models such as Logistic Regression, Naïve Bayes, and Support Vector Machines have been widely used for text classification due to their interpretability and efficiency [15]. These models are particularly suitable for safety-critical systems where explainability is essential.

Recent studies have explored deep learning models including CNNs, RNNs, and transformer-based architectures for richer semantic understanding [16]. Sentence embedding models have demonstrated superior performance in topic classification tasks involving short and noisy text, such as video titles and descriptions.

2.9 JUSTIFICATION OF MACHINE LEARNING TECHNIQUES USED

Based on insights from the reviewed literature, the proposed system adopts a hybrid machine learning architecture. Logistic Regression combined with TF-IDF features is used for age classification and safety screening, as supported by prior research emphasizing interpretability and reliability [15]. Confidence thresholds and rejection mechanisms further enhance safety.

For topic classification, deep neural network–based sentence embeddings are employed to capture semantic nuances, aligning with findings from embedding-based classification studies [16]. This combination ensures a balance between accuracy, explainability, computational efficiency, and child safety, addressing limitations identified in existing approaches.

The effectiveness of an intelligent child-oriented video filtering and recommendation system depends heavily on the quality, diversity, and relevance of the datasets used for training, validation, and evaluation. This project utilizes a combination of child-language corpora, age-labeled text datasets, harmful-content datasets, and real-world YouTube metadata, each selected to support a specific component of the system such as age classification, safety analysis, topic classification, and recommendation filtering.

The CHILDES (Child Language Data Exchange System) corpus, developed by MacWhinney et al. under the TalkBank project, provides transcribed child-directed and child-produced speech annotated with age information. These datasets capture linguistic patterns such as repetition, vocabulary simplicity, and syntactic development across different childhood stages.

The TextAge dataset, introduced by Xia et al., is a benchmark dataset designed for age prediction from text. It includes age-labeled textual samples that highlight differences in lexical complexity and semantic structure across age groups.

Curated datasets consisting of children’s poems (ages 6–12) and spoken child sentences (ages 4–5) were used to train supervised age classifiers. Poems provided structured, age-targeted language suitable for distinguishing early-school and middle-school content, while spoken child datasets captured informal and repetitive speech patterns characteristic of preschool-aged children.

The Elsagate dataset, analyzed by Papadamou et al., contains examples of videos that appear child-friendly but include disturbing or inappropriate themes. Large-scale datasets such as YouTube-8M and the IN YouTube Trending Dataset were used to study real-world video metadata, category distributions, and topic diversity.

To obtain textual representations of video content, automatic speech recognition transcripts generated using the Whisper model were employed. The Children’s Online Privacy Protection Act (COPPA) documentation provided regulatory guidance rather than training data. It informed dataset usage policies, age segmentation decisions, and the inclusion of parental controls to ensure ethical and legal compliance.

TABLE 2.10 DATASETS USED

S.NO	DATASET	DESCRIPTION	URL
1	TextAge: A Curated and Diverse Text Dataset for Age Classification	TextAge is a curated dataset designed for predicting age groups based solely on textual content. It contains age-labeled text samples that capture variations in vocabulary complexity, sentence structure, repetition, and semantic richness.	https://arxiv.org/html/2406.16890v1
2	Disturbed Youtube for Kids	The Disturbed YouTube for Kids dataset contains examples of YouTube videos that are deceptively presented as child-friendly but include disturbing, inappropriate, or harmful themes. The dataset was introduced to study the Elsagate phenomenon and analyze patterns of unsafe children’s content propagated through recommendation algorithms.	https://github.com/kwstantos-papadamou/disturbed-youtube-for-kids

			<u>classifier/tree/master</u>
3	In Youtube Trending Videos	The IN YouTube Trending Videos dataset is a region-specific collection of trending YouTube videos in India, containing metadata such as titles, descriptions, tags, category IDs, and engagement metrics.	<u>https://www.kaggle.com/datasets/rsrishav/youtube-trending-video-dataset</u>
4	YouTube-8M	YouTube-8M is a large-scale labeled video dataset released by Google Research, containing millions of YouTube video IDs with pre-extracted visual and audio features.	<u>https://research.google.com/youtube8m/</u>

CHAPTER 3

SYSTEM ARCHITECTURE

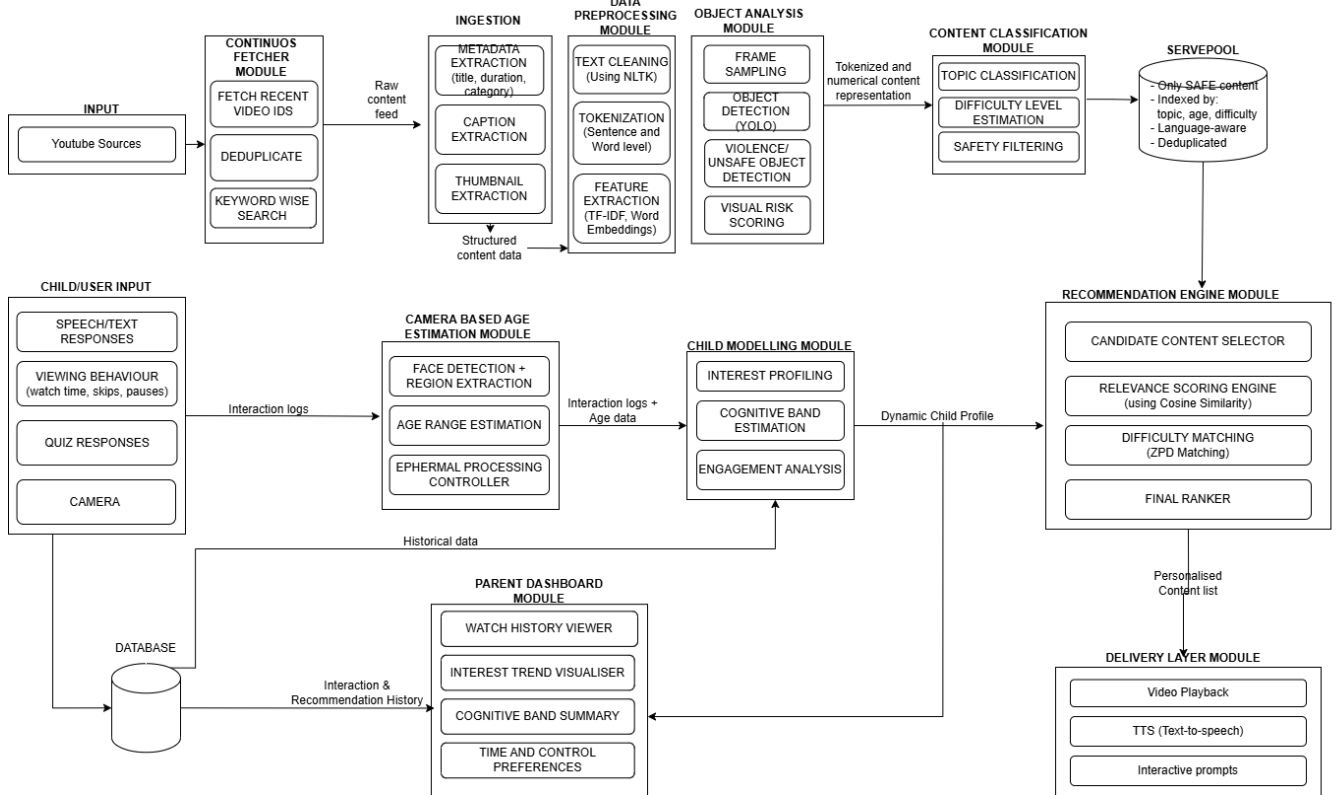
3.1 OVERVIEW

The proposed system is a multimodal, safety-aware, cognitively adaptive video recommendation architecture designed for child-centric educational content delivery. The system integrates content ingestion, multimodal analysis, safety filtering, supervised difficulty estimation, user cognitive modeling, and adaptive recommendation ranking.

The architecture consists of six major layers:

1. Content Acquisition Layer
2. Content Analysis Layer (Text + Vision)
3. Safety & Classification Layer
4. Safe Content Repository
5. User Modeling Layer
6. Recommendation & Delivery Layer

Each module in the architecture diagram is explained below.



3.2 CONTENT ACQUISITION LAYER

3.2.1 YOUTUBE SOURCES

This module acts as the primary source of video content by retrieving educational and child-appropriate videos from YouTube using API-based keyword queries. Leveraging YouTube ensures access to a large and continuously updated pool of real-world content, allowing the system to operate on diverse and relevant data without requiring manual dataset creation.

3.2.2 CONTINUOUS FETCHER MODULE

The Continuous Fetcher Module enables dynamic and incremental content ingestion, ensuring that the dataset remains current over time. It retrieves recently uploaded videos as well as performs keyword-based searches to expand the content pool. This approach prevents the system from becoming outdated and allows it to scale efficiently as new content becomes available.

3.3 METADATA & RAW CONTENT EXTRACTION

3.3.1 METADATA EXTRACTION

Metadata such as title, duration, and category is extracted from each video to provide initial contextual information. These attributes offer lightweight yet valuable signals that assist in early-stage filtering, topic classification, and estimation of content complexity.

3.3.2 TRANSCRIPT EXTRACTION

Subtitles or auto-generated captions are retrieved using OpenAI Whisper STT and used as the primary input for text-based analysis. Since captions represent the spoken content of the video, they enable semantic understanding without requiring full video processing, making the system more efficient.

3.4 TEXT PROCESSING PIPELINE

3.4.1 TEXT CLEANING (USING NLTK)

The extracted caption text is preprocessed by removing stop words, special characters, and noise tokens to ensure clean and normalized input. This step improves the quality of downstream analysis by eliminating irrelevant information.

3.4.2 TOKENIZATION

The cleaned text is divided into sentences and words, enabling detailed linguistic analysis. This allows the system to examine structure, word distribution, and complexity, which are important for classification and difficulty estimation.

3.4.3 FEATURE EXTRACTION

Text is transformed into numerical representations using TF-IDF vectors and word embeddings. While TF-IDF captures the importance of words within the dataset, embeddings provide semantic context, allowing the system to better understand meaning. These features are used for topic classification, relevance scoring, and estimating the difficulty level of content.

3.5 VISION PROCESSING PIPELINE

3.5.1 FRAME SAMPLING

Instead of processing every frame, representative frames are extracted at fixed intervals to reduce computational cost while maintaining sufficient visual coverage of the video.

3.5.2 OBJECT DETECTION (YOLO)

A pre-trained YOLO model is used to detect objects present in the sampled frames. Its real-time detection capability makes it suitable for efficiently identifying relevant visual elements.

The detected objects are analyzed to identify potentially harmful or inappropriate elements such as weapons or explicit visuals, which may not be captured through text alone.

A visual risk score is computed based on the frequency and severity of unsafe detections across frames. This scoring approach allows for a more nuanced evaluation of content safety rather than relying on binary decisions.

3.6 SAFETY AND CLASSIFICATION LAYER

3.6.1 TOPIC CLASSIFICATION

Using features derived from text processing, videos are classified into educational domains such as mathematics, science, or language. This categorization helps organize content and ensures alignment with user interests.

3.6.2 DIFFICULTY LEVEL ESTIMATION

A difficulty score is assigned by analyzing vocabulary complexity, sentence structure, and conceptual density. This enables the system to match content with the user's learning level.

3.6.3 SAFETY FILTERING

Outputs from both text and vision pipelines are combined to evaluate overall content safety. Only videos that satisfy predefined thresholds are allowed into the next stage, ensuring that unsafe content is excluded early.

3.7 SAFE CONTENT REPOSITORY

The repository stores only approved and filtered videos, organized by topic, difficulty, and language, with duplicate entries removed. This separation ensures that only safe and relevant content is accessible during recommendation, improving reliability and response time.

3.8 USER MODELING LAYER

3.8.1 INPUT SIGNALS

The system collects various interaction signals including viewing behavior such as watch time and skips, quiz performance, and camera input, enabling a comprehensive understanding of the user.

3.8.2 AGE RANGE ESTIMATION

An approximate age range is predicted using a trained model supported by CUDA and benchmark datasets, which helps initialize difficulty alignment for new users.

3.9 RECOMMENDATION ENGINE

3.9.1 CANDIDATE CONTENT SELECTOR

A subset of videos is retrieved from the Safe Content Repository based on topic relevance, reducing the search space for recommendation.

Cosine similarity is used to measure the similarity between user interest vectors and video feature vectors, enabling efficient identification of semantically relevant content.

3.9.3 DIFFICULTY MATCHING (ZPD MATCHING)

The system applies Zone of Proximal Development principles by selecting videos with difficulty slightly above the user's current cognitive level, promoting effective learning without overwhelming the user.

3.9.4 FINAL RANKER

The final ranking combines relevance, difficulty alignment, engagement history, and novelty to produce a balanced and personalized recommendation list.

3.10 USER INTERFACE AND DELIVERY LAYER

3.10.1 WATCH HISTORY VIEWER

Displays previously watched videos along with behavioral insights, helping track user interaction.

3.10.2 INTEREST TREND VISUALISER

Represents how user interests evolve over time across different topics.

3.10.3 COGNITIVE BAND SUMMARY

Provides an interpretable view of the user's learning progress.

3.10.4 TIME AND CONTROL PREFERENCES

Allows implementation of parental controls such as usage limits and content restrictions.

3.10.5 VIDEO PLAYBACK MODULE

Handles streaming and playback of recommended videos within the system.

3.10.6 TEXT-TO-SPEECH (TTS)

Provides optional audio narration support, improving accessibility.

3.11 END-TO-END FLOW

The system follows a complete pipeline where videos are continuously fetched, processed through metadata, text, and visual analysis, and filtered based on safety constraints before being stored. User interaction signals are then used to model cognitive ability and preferences, enabling the recommendation engine to generate personalized and adaptive video suggestions. Continuous feedback from user behavior updates the model over time, ensuring that recommendations remain relevant and aligned with the user's learning progression.

CHAPTER 4

IMPLEMENTATION OF THE SYSTEM

4.1 OVERALL SYSTEM ARCHITECTURE

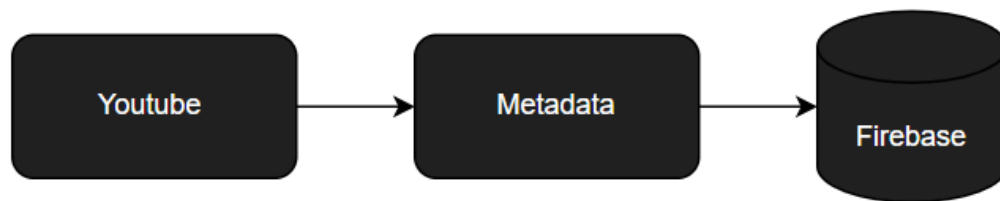
The overall architecture of the system integrates multiple modules including data ingestion, preprocessing, age classification, safety analysis, recommendation engine, and delivery interface.

The system processes video data through sequential stages, transforming raw input into personalized recommendations for children.

4.2 DATA INGESTION AND STORAGE

This module handles the collection and storage of video data from external sources such as YouTube.

Figure 4.2 shows the ingestion pipeline where video metadata is extracted and stored in Firebase.



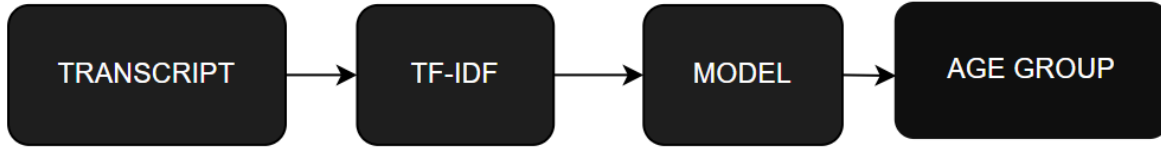
Explanation:

As shown in Figure 4.2, the system extracts video metadata including title, transcript, and identifiers. This structured data is stored in Firebase Firestore, enabling efficient retrieval and scalability for large datasets.

4.3 TEXT PREPROCESSING AND AGE CLASSIFICATION

The transcript of each video is processed and classified into an age group using a machine learning model.

Figure 4.3 illustrates the pipeline for text preprocessing and age classification.



Explanation:

Figure 4.3 depicts how raw transcript data is transformed into numerical vectors using TF-IDF vectorization. These vectors are fed into a trained classification model, which predicts the appropriate age group for the video. This classification ensures that only age-appropriate content is recommended.

4.4 SAFETY ANALYSIS USING OBJECT DETECTION

This module evaluates the safety of video content using computer vision techniques.

Figure 4.4 shows the safety analysis workflow.



Explanation:

As illustrated in Figure 4.4, frames are extracted from videos and processed using a YOLO-based object detection model. The presence of unsafe objects contributes to a computed risk score. This score is used to filter or penalize unsafe content in the recommendation process.

4.5 CHILD PROFILE MODELLING

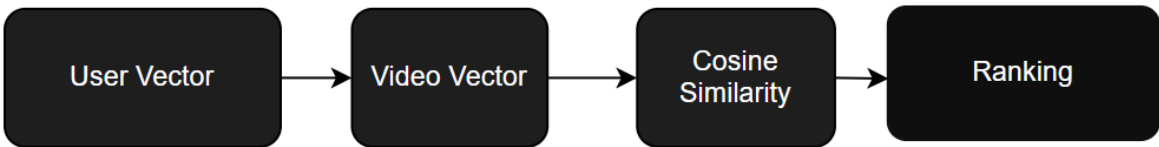
The system builds a dynamic profile of the child based on interaction history.

User interactions such as watched and skipped videos are used to construct a preference vector. This vector captures the interests of the child and evolves over time, enabling personalized recommendations.

4.6 VECTOR-BASED RECOMMENDATION ENGINE

The recommendation engine ranks videos using vector similarity techniques.

Figure 4.6 illustrates the recommendation workflow.



Explanation:

As shown in Figure 4.6, both user preferences and video features are represented as vectors. Cosine similarity is used to measure the relevance between them. Videos are filtered based on age and safety constraints, and then ranked according to similarity scores. Additional adjustments are made using watch history to improve personalization.

4.7 VIDEO DELIVERY AND USER INTERFACE

The final module delivers recommended content through a mobile interface.

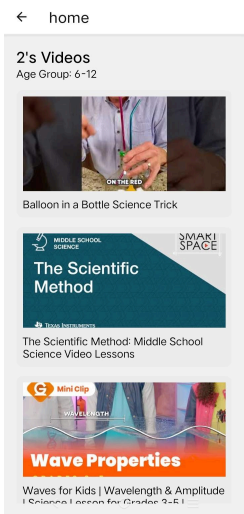


Figure 4.7 shows the player interface with recommendations.

Explanation:

Figure 4.7 presents the user interface of the system. The selected video is played using a YouTube player, and recommended videos are displayed below. Users can seamlessly switch between videos, and the recommendations update dynamically based on interactions.

CHAPTER 5

IMPLEMENTATION

This chapter details the technical implementation of the AI-driven child-safe video recommendation system. The system execution is divided into multiple programmatic modules, each responsible for handling specific functional tasks such as facial age classification, dataset balancing, model training, recommendation generation, and ONNX-based inference.

5.1 FACIAL AGE CLASSIFICATION ALGORITHM

To ensure that children are presented with age-appropriate content, the system employs a deep learning-based facial age classification algorithm. The model processes an input image, extracts relevant features and predicts the appropriate age group.

The classification is performed across three categories: 1–3, 4–5, and 6–12.

Algorithm 5.1 Facial Age Classification

```
1: Input: Face Image (I)
2: Output: Predicted Age Group (A)
3: Resize image I to  $224 \times 224$ 
4: Normalize pixel values of I
5: Convert image to tensor format
6: Pass tensor through trained CNN model
7: Obtain output logits
8:  $\text{predicted\_class} \leftarrow \text{ArgMax}(\text{logits})$ 
9: if  $\text{predicted\_class} == 0$  then
10:    $A \leftarrow \text{"1-3"}$ 
11: else if  $\text{predicted\_class} == 1$  then
12:    $A \leftarrow \text{"4-5"}$ 
13: else if  $\text{predicted\_class} == 2$  then
14:    $A \leftarrow \text{"6-12"}$ 
15: else
16:    $A \leftarrow \text{"13+"}$ 
17: end if
18: return
```

5.2 MODEL TRAINING PROCEDURE

The system trains a convolutional neural network using a supervised learning approach. The training process involves forward propagation, loss computation, backpropagation, and weight updates across multiple epochs.

Algorithm 5.3 Model Training Procedure

```
1: Input: Dataset (D), Model (M), Number of Epochs (E)
2: Output: Trained Model (M*)
3: Initialize optimizer and loss function
4: for epoch  $\leftarrow$  1 to E do
5:   total_loss  $\leftarrow$  0
6:   correct  $\leftarrow$  0
7:   total  $\leftarrow$  0
8:   for each batch (X, Y) in D do
9:     outputs  $\leftarrow$  M.forward(X)
10:    loss  $\leftarrow$  ComputeLoss(outputs, Y)
11:    Perform Backpropagation(loss)
12:    Update model weights
13:    total_loss  $\leftarrow$  total_loss + loss
14:    predictions  $\leftarrow$  ArgMax(outputs)
15:    correct  $\leftarrow$  correct + Count(predictions == Y)
16:    total  $\leftarrow$  total + Size(Y)
17:   end for
18:   accuracy  $\leftarrow$  correct / total
19:   Print(epoch, total_loss, accuracy)
20: end for
21: Save trained model
22: return M*
```

5.3 VIDEO RECOMMENDATION ALGORITHM

The recommendation engine gives personalized video suggestions based on user preferences, safety constraints, and content similarity. It uses vector-based similarity techniques to rank videos.

Algorithm 5.4 Video Recommendation Engine

```
1: Input: User Profile (U), Video Dataset (V), Threshold (T)
2: Output: Ranked Recommendation List (R)
3: R  $\leftarrow$  Empty List
4: for each video v in V do
5:   if v.age_group  $\neq$  U.age_group then
6:     continue (Hard Filter: Age-Appropriateness)
7:   end if
8:   if v.safety_score < T then
9:     continue (Hard Filter: Safety/Risk Mitigation)
10:  end if
11:  video_vector  $\leftarrow$  Extract latent features from v (Topic, Metadata)
12:  user_vector  $\leftarrow$  Extract semantic preferences from U (History, Enriched Topics)
13:  similarity  $\leftarrow$  CosineSimilarity(user_vector, video_vector)
14:  if v exists in user watch history then
15:    if v was skipped then
16:      similarity  $\leftarrow$  similarity  $\times$  0.6 (Negative Feedback Penalty)
17:    else if v was watched then
18:      similarity  $\leftarrow$  similarity  $\times$  1.2 (Positive Reinforcement)
19:    end if
20:  end if
21:  Add (v, similarity) to R
22: end for
23: Sort R in descending order by similarity coefficient
24: R  $\leftarrow$  Select top-k videos (Final Ranking)
25: return R
```

5.4 ONNX MODEL INFERENCE ALGORITHM

To enable efficient deployment, the trained PyTorch model is converted into ONNX format. The ONNX model is used for fast inference in the backend system.

Algorithm 5.5 ONNX-Based Age Prediction

```
1: Input: Image (I)
2: Output: Predicted Age Group (A)
3: Resize image I to  $224 \times 224$ 
4: Normalize pixel values
5: Convert image to array format
6: Transpose array to (C, H, W)
7: Expand dimensions to create batch
8: output  $\leftarrow$  Run ONNX inference
9: predicted_class  $\leftarrow$  ArgMax(output)
10: if predicted_class == 0 then
11:   A  $\leftarrow$  "1-3"
12: else if predicted_class == 1 then
13:   A  $\leftarrow$  "4-5"
14: else
15:   A  $\leftarrow$  "6-12"
16: end if
17: return A
```

CHAPTER 6

RESULTS AND PERFORMANCE ANALYSIS

6.1 RESULTS AND INTERFACES

This section presents the outputs generated by the system along with the user interface developed for interaction. The results demonstrate the effectiveness of the system in delivering age-appropriate, safe, and personalized video recommendations.

The system integrates multiple modules including age classification, safety analysis, and recommendation ranking, and the outputs of each module are reflected in the final user interface.

6.1.1 AGE AND TOPIC CLASSIFIED VIDEO LISTING

The home screen displays videos filtered based on the detected age group of the child.

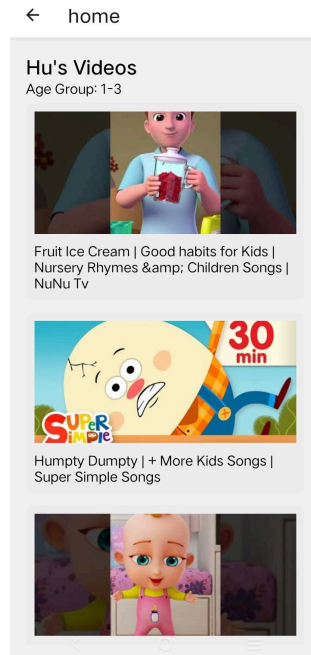


Figure 6.1 shows the home screen with filtered videos.

Explanation

As shown in Figure 6.1, the system displays videos corresponding to the detected age group. The age group is obtained either from the camera-based detection module or from default selection.

Videos are classified based on age - '1-3', '4-5' '6-12' and Topics according to the Youtube 8M dataset. Finally, video recommendations of that age and their interested topic is provided.

Only videos matching the selected age group are retrieved from the database. This ensures that inappropriate content is not shown to the child.

The interface displays:

- Video thumbnails
- Titles
- Clickable video cards

This filtering mechanism significantly improves content safety.

6.1.2 VIDEO PLAYER WITH RECOMMENDATIONS

The player interface allows users to watch videos and view recommended content.

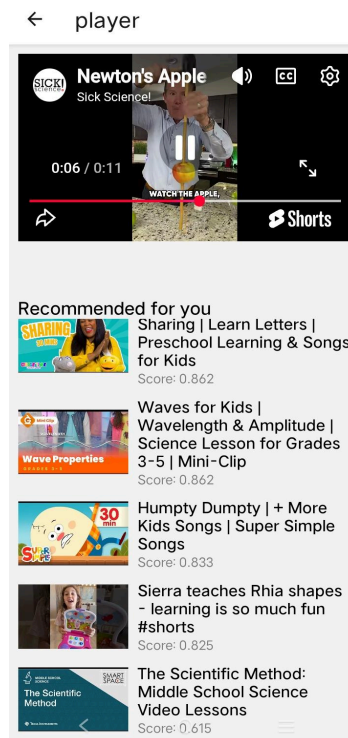


Figure 6.2 shows the video player along with recommended videos.

Explanation

Figure 6.2 illustrates the video playback interface. The selected video is played using a YouTube iframe player.

Below the player, a list of recommended videos is displayed. These recommendations are generated dynamically based on:

- User preference vector
- Video feature similarity
- Safety constraints

When a user selects a recommended video, the player updates instantly and new recommendations are generated.

6.1.3 AGE DETECTION OUTPUT

The system includes an age detection module that determines the child's age group.

INPUT:



OUTPUT:

```
Response body
{
  "age_group": "6-12"
}

Response headers
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:5000
content-length: 20
content-type: application/json
date: Sat, 04 Apr 2026 12:15:10 GMT
server: uvicorn
vary: Origin
```

Fig. 6.3 shows the server getting a sample image as input and predicting the age of the person

Explanation

As shown in Figure 6.3, the system returns an age group such as “1–3”, “4-5”, “6–12” or “13+” based on the input image or transcript.

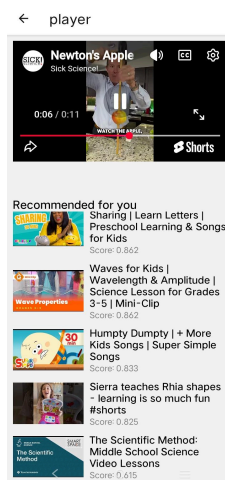
This age group acts as a strict filter in the recommendation pipeline. Only videos belonging to the same age group are considered for ranking. The integration of age detection ensures

personalized and safe content delivery. A model was trained with a ground truth dataset containing images of ages 1-100.

6.1.4 RECOMMENDATION OUTPUT VISUALIZATION

This section presents the results generated by the recommendation engine.

Figure 6.4 shows the recommended video list.



Explanation

Figure 6.4 displays the top-ranked videos generated by the recommendation engine.

The ranking is based on cosine similarity between:

- User preference vector
- Video feature vectors

Each video is assigned a score, and higher scoring videos are displayed first.

The system ensures:

- Age consistency
- Safety filtering
- Personalized ranking

6.2 PERFORMANCE ANALYSIS

This section evaluates the performance of different modules in the system.

6.2.1 AGE DETECTION MODEL PERFORMANCE ANALYSIS

This section evaluates the performance of the proposed age detection model trained using a deep learning approach in PyTorch. The model is assessed using training metrics across epochs and classification metrics on the test dataset to demonstrate its effectiveness.

The age detection model was trained for 10 epochs, and its performance was monitored using loss and accuracy metrics.

Table 6.1: Training Performance Summary

Epoch	Loss	Accuracy
1	0.2460	81.94%
2	0.0892	93.19%
3	0.0971	93.20%
4	0.0432	97.11%
5	0.0605	95.71%
6	0.0471	97.03%
7	0.0585	95.40%
8	0.0505	96.26%
9	0.0152	98.38%
10	0.0261	98.55%

As shown in Table 6.1, the model demonstrates rapid convergence, with accuracy increasing from 81.94% in the first epoch to 98.55% in the final epoch. The training loss decreases significantly, indicating effective learning of age-related features. Minor fluctuations in loss during intermediate epochs suggest regularization effects and improved generalization rather than overfitting.

Classification Performance

The trained model was evaluated on a test dataset of 7,778 samples across three age classes using precision, recall, and F1-score metrics.

Table 6.2: Classification Report

Class	Precision	Recall	F1-score	Support
0	0.99	1.00	0.99	2572
1	0.96	1.00	0.98	2597
2	1.00	0.95	0.97	2609

Overall Accuracy: 98%

Macro Average F1-score: 0.98

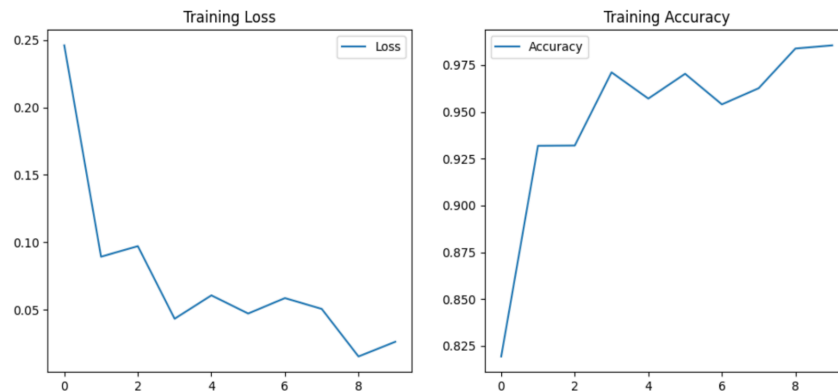
Weighted Average F1-score: 0.98

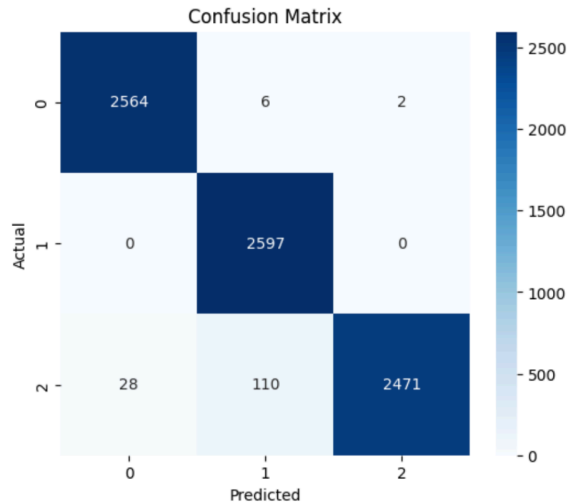
The results indicate strong classification performance across all classes. Class 0 achieves near-perfect scores, while Class 1 shows excellent recall with slightly lower precision. Class 2 achieves perfect precision but slightly reduced recall, indicating minimal false positives but a few missed predictions.

Performance Discussion

The model achieves an overall accuracy of 98%, demonstrating high reliability in age classification tasks. The balanced macro and weighted averages confirm that the model performs consistently across all classes without bias.

The high recall values for Classes 0 and 1 indicate effective detection capability, while the high precision for Class 2 reflects accurate classification with minimal mislabeling. These results suggest that the model generalizes well and is suitable for real-time deployment in applications such as surveillance and user profiling.





The confusion matrix illustrates the performance of the trained age classification model across three classes: 1–3 (class 0), 4–5 (class 1), and 6–12 (class 2). The diagonal elements represent correct predictions, while off-diagonal values indicate misclassifications.

Overall, the confusion matrix indicates that the model has high accuracy and strong class separation, particularly for the first two age groups, while minor misclassification occurs primarily within neighboring age ranges.

6.2.2 SAFETY ANALYSIS PERFORMANCE

The safety module is evaluated based on its ability to detect unsafe content.

Observation

- YOLO detects objects in video frames effectively
- Risk score helps filter unsafe videos
- Performance depends on frame quality

The safety filtering mechanism improves content reliability by reducing exposure to harmful content. However, false positives may occur in complex scenes.

6.2.3 SYSTEM EFFICIENCY

The system performance is analyzed in terms of responsiveness and execution time.

Observation

- Firebase queries are fast
- UI updates are smooth
- Recommendation computation is efficient

The use of vector-based similarity ensures low computational overhead. The system can handle real-time recommendation updates without noticeable delay.

6.3 LIMITATIONS

Despite successful implementation, the system has certain limitations:

- Dependence on transcript quality for age classification
- Network issues affecting API communication (e.g., fetch errors)
- Limited training data for recommendation accuracy

6.4 SUMMARY

This chapter presented the results and performance evaluation of the system. The implemented modules successfully generate age-appropriate and personalized recommendations.

The integration of age classification, safety analysis, and vector-based recommendation improves both relevance and safety of content. The system demonstrates good performance in real-time scenarios, with scope for further optimization and enhancement.

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